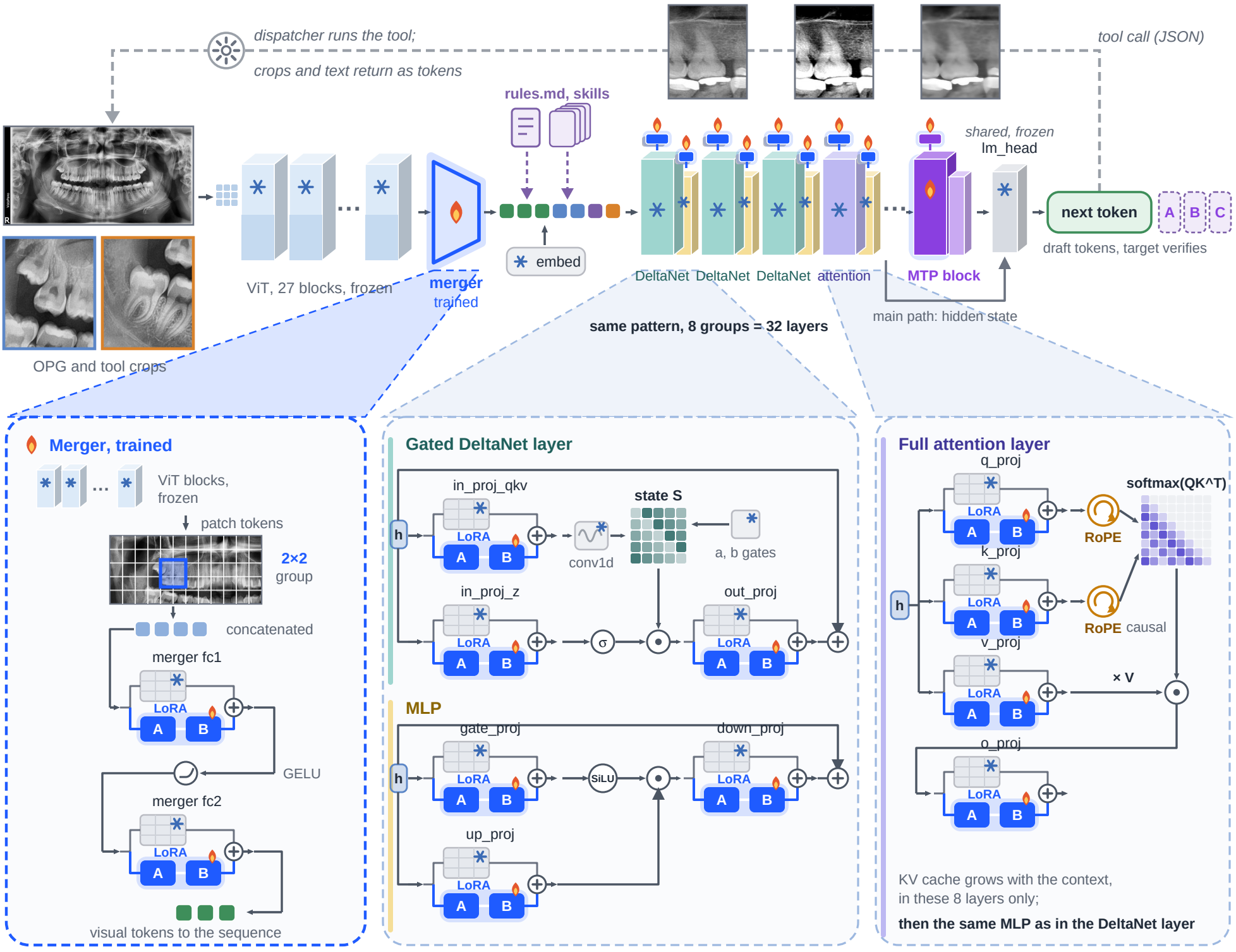

dispatcher runs the tool;
crops and text return as tokens
tool call (JSON)
rules.md, skills
shared, frozen
lm_head
next token
A B C
draft tokens, target verifies
embed
ViT, 27 blocks, frozen
merger
trained
DeltaNet DeltaNet DeltaNet attention
MTP block
main path: hidden state
same pattern, 8 groups = 32 layers
OPG and tool crops
Merger, trained
ViT blocks, frozen
patch tokens
2×2 group
concatenated
merger fc1
LoRA
A B
GELU
merger fc2
visual tokens to the sequence
Gated DeltaNet layer
in_proj_qkv
state S
conv1d
a, b gates
in_proj_z
out_proj
MLP
gate_proj
down_proj
SiLU
up_proj
Full attention layer
q_proj
softmax(QK^T)
RoPE
k_proj
RoPE causal
v_proj
× V
o_proj
KV cache grows with the context,
in these 8 layers only;
then the same MLP as in the DeltaNet layer
frozen
trained
LoRA: GRPO adapter in front of the frozen SFT adapter (KL reference)
MTP block, fine-tuned
frozen: 4-bit base weights, ViT blocks, embeddings, lm_head, norms, conv1d, a/b gates
trained: LoRA on q, k, v, o, in_proj_qkv, in_proj_z, out_proj, gate, up, down; the two merger layers; the MTP block